

x-up-devcap-post-charset Header in ASP.NET to Bypass WAFs Again!

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- Published: date not stated
- Original: <<https://soroush.me/blog/x-up-devcap-post-charset-header-in-aspnet-to-bypass-wafs-again>>
- Preserved from: <https://soroush.me/blog/x-up-devcap-post-charset-header-in-aspnet-to-bypass-wafs-again> (live) on 2026-08-06
- Licence: unknown

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x-up-devcap-post-charset Header in ASP.NET to Bypass WAFs Again!

In the past, I showed how the request encoding technique can be abused to bypass web application firewalls (WAFs). The generic WAF solution to stop this technique has been implemented by only allowing whitelisted `charset` via the `Content-Type` header or by blocking certain encoding charsets. Although WAF protection mechanisms can normally be bypassed by changing the headers slightly, I have also found a new header in ASP.NET that can hold the `charset` value which should bypass any existing protection mechanism using the `Content-Type` header.

Let me introduce to you, the one and only, the `x-up-devcap-post-charset` header that can be used like this:

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```
1POST /test/a.aspx?%C8%85%93%93%96%E6%96%99%93%84= HTTP/1.1
```

```
2Host: target
```

```
3User-Agent: UP foobar
```

```
4Content-Type: application/x-www-form-urlencoded
```

```
5x-up-devcap-post-charset: ibm500
```

```
6Content-Length: 40
```

```
7
```

```
8%89%95%97%A4%A3%F1=%A7%A7%A7%A7%A7%A7%A7
```

As it is shown above, the `Content-Type` header does not have the `charset` directive and the `x-up-devcap-post-charset` header holds the encoding's charset instead. In order to tell ASP.NET to use this new header, the `User-Agent` header should start with `UP !`

The parameters in the above request were create by the [Burp Suite HTTP Smuggler](#), and this request is equal to:

Copy

```
1POST /testme87/a.aspx?HelloWorld= HTTP/1.1
```

```
2Host: target
```

```
3User-Agent: UP foobar
```

```
4Content-Type: application/x-www-form-urlencoded
```

```
5Content-Length: 14
```

```
6
```

```
7input1=xxxxxxx
```

I found this header whilst I was looking for something else inside the ASP.NET Framework. Here is how ASP.NET reads the content encoding before it looks at the `charset` directive in the `Content-Type` header:

<https://github.com/Microsoft/referencesource/blob/3b1eaf5203992df69de44c783a3eda37d3d4cd10/System/net/System/Net/HttpListenerRequest.cs#L362>

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```
1 if (UserAgent!=null && CultureInfo.InvariantCulture.CompareInfo.IsPrefix(UserAgent, "UP")) {  
  
2     string postDataCharset = Headers["x-up-devcap-post-charset"];  
  
3     if (postDataCharset!=null && postDataCharset.Length>0) {  
  
4         try {  
  
5             return Encoding.GetEncoding(postDataCharset);
```

Or

<https://github.com/Microsoft/referencesource/blob/08b84d13e81cfd769a557b368539aac6a9cb30/System.Web/HttpRequest.cs#L905>

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```
1 if (UserAgent != null && CultureInfo.InvariantCulture.CompareInfo.IsPrefix(UserAgent, "UP")) {  
  
2     String postDataCharset = Headers["x-up-devcap-post-charset"];  
  
3     if (!String.IsNullOrEmpty(postDataCharset)) {  
  
4         try {  
  
5             return Encoding.GetEncoding(postDataCharset);
```

I should admit that the original technique still works on most of the WAFs out there as they have not taken the request encoding bypass technique seriously ;) However, the OWASP ModSecurity Core Rule Set (CRS) quickly created a simple rule for it at the time which they are going to improve in the future. Therefore, I disclosed this new header to Christian Folini (@ChrFolini) from CRS to create another useful rule before releasing this blog post. The pull request for the new rule is pending at <https://github.com/SpiderLabs/owasp-modsecurity-crs/pull/1392>.

**References:* <https://soroush.me/downloadable/request-encoding-to-bypass-web-application-firewalls.pdf> <https://www.slideshare.net/SoroushDalili/waf-bypass-techniques-using-http-standard-and-web-servers-behaviour> <https://soroush.secproject.com/blog/2018/08/waf-bypass-techniques-using-http-standard-and-web-servers-behaviour/> https://soroush.me/downloadable/Rare_ASP.NET_Request_Validation_Bypass_Using_Request-Encoding.pdf <https://github.com/nccgroup/BurpSuiteHTTPSmuggler/>

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Creation date: May 4, 2019

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